PHY II – 302

Waves, Oscillations & Thermodynamics
Morning 2026
Institute of Space Science & Technology
University of Karachi

Beyond the Spring

Oscillation as the Universal Language of Nature

TOTAL MARKS	10
ASSIGNMENT NO.	I
DUE DATE	12, 10/26
SUBMISSION MODE	Typed .docx named *"PHY_II_Assignment_I_StudentName"*
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Academic Integrity & Grading Policy

1. ***Mutual assistance is encouraged.*** *Students are allowed to discuss the topics, share thoughts, and take references from the seniors or anyone reliable including usage of AI only to grab ideas. This will do no harm to your grading. However:*
2. ***Original work only.*** *Every response must be written in your own words and reasoning. Copying from the internet, a textbook, a classmate, or an AI tool's output is not permitted anywhere in this assignment. You are encouraged to research using credible sources (this is expected for Part A), but the explanation must be reconstructed in your own understanding, with sources credited.*
3. ***Similarity penalty.*** *If two or more submissions are found to contain matching or substantially unparaphrased content, every submission in that matching group is penalized. The penalty to each submission equals the size of the matching group:*

Matching submissions found	*Penalty to each submission*
2 submissions match	*–4 marks*
3 submissions match	*–6 marks*
4 submissions match	*–8 marks*
n submissions match	*–2n marks*

4. ***Relevant to Final Exam.*** *This assignment will be evaluated in connection with the conduct of the final examination, based on the relevant questions included in your exam paper, in the end.*

This matters beyond this assignment. The reasoning built here – recognizing an oscillation hiding inside an unfamiliar system – is a core, examinable learning outcome for this unit. Related questions may reappear in quizzes, in-class discussion, and the final examination.

What This Assignment Asks of You

- Recognize simple harmonic motion as a mathematical pattern, not a spring-specific phenomenon.
- Connect Lecture derivation to real engineering systems and to frontier physics research.
- Apply the SHM toolkit – ω, amplitude, initial conditions to an unfamiliar physical system.
- Practice communicating a technical idea precisely, in your own words, with credible support.

Before You Begin

Every machine, every signal, and perhaps every particle in the universe owes its behaviour to the same mathematical idea you derived in your lectures: a system displaced from equilibrium, pulled back by some restoring influence, tracing out sine and cosine in time.

Start close to home. The engine in a car converts the linear push of a piston into rotation through a crankshaft – quite literally the linear-motion-as-a-projection-of-rotation relationship you met when asking why a bouncing spring knows about circles. Zoom out to the frontier of physics: the Higgs boson was not seen the way a photograph sees an object – it was discovered as a resonant bump in collision-energy data at the Large Hadron Collider, the signature of the Higgs field itself being excited into oscillation above its ground state. Zoom out again: in 2015, LIGO detected gravitational waves for the first time by measuring an oscillation in spacetime itself – space stretching and compressing by a fraction of the width of a proton, a ripple sent outward as two black holes spiralled into each other over a billion years ago.

Now zoom back to your own senses. You hear because air molecules oscillate, carrying pressure disturbances to your eardrum. You see because electric and magnetic fields oscillate together, endlessly regenerating one another as they race outward at the speed of light. In every one of these cases – a piston, a quantum field, the fabric of space, the air, light itself – some quantity is oscillating. It is the question "what, exactly, is oscillating here?" that has repeatedly led physicists to their deepest discoveries.

PART A: Conceptual & Research-Based [5 Marks]

Question 1: Choose ONE [2 Marks]

Choose one of the following:

(a) the discovery of the Higgs boson, or

(b) the LIGO detection of gravitational waves.

Using at least one credible, cited source, explain in your own words (100 ≥ words):

(i) what physical quantity is oscillating; and

(ii) how a concept from the lectures; amplitude, frequency, or a restoring influence shows up, even loosely, in how the phenomenon is described or detected.

Question 2: Choose ONE [2 Marks]

Choose one of the following:

(a) AC electricity generation, or

(b) the orbital motion of a satellite or planet.

In 100 ≥ words, and using your own reasoning, explain how the same angular frequency ω that appeared in your spring-mass solution also governs your chosen system and what physical rotation is "hiding behind" the linear or periodic behaviour you observe.

Question 3 [1 Mark]

In 3–4 sentences: why do physicists so often ask *"what is oscillating?"* when they encounter a new wave-like phenomenon (sound, light, gravitational waves, a quantum field)? What does correctly identifying the oscillating quantity let them go on to predict or calculate?

PART B: Numerical & Analytical [4 Marks]

Question 4: Spring–Mass System [2 Marks]

A block of mass $M = 0.50\ kg$ is attached to a horizontal spring of stiffness $k = 200\ N/m$ on a frictionless surface. At $t = 0$ it is pulled to $x = 4\ cm$ from equilibrium and released from rest.

(a) Find the angular frequency ω.

(b) Write $x(t)$.

(c) Find the period T.

(d) Find the block's maximum speed.

(e) Find the block's maximum acceleration.

Question 5: The Engine, Revisited [2 Marks]

An idealized engine piston moves in simple harmonic motion as its crankshaft turns at a constant angular speed. (Model this as a Scotch-yoke mechanism – ignore the finite length of the connecting rod, which real engines must account for.) The engine runs at $N = 3000\ rpm$, and the piston's stroke (its full peak-to-peak travel) is $8\ cm$. Take $t = 0$ at the instant the piston is at top dead centre – maximum displacement, momentarily at rest.

(a) Convert the engine's rotational speed to an angular frequency ω, in rad/s.

(b) Write *x(t)* for the piston's position, measured from the midpoint of its stroke.

(c) Find the piston's maximum speed.

(d) Find the piston's maximum acceleration, expressed as a multiple of g = 9.81 m/s^2. In one sentence, comment on what this implies for how engine components must be designed.

PART C: Synthesis [1 Mark]

Question 6 [1 Mark]

Identify one more real-world or natural phenomenon – not discussed in Lectures or elsewhere in this assignment – that is, at its core, an oscillation. In 2–3 sentences, state what quantity is oscillating, and why recognizing it as an oscillation is useful.

Grading Rubric

Criterion	Weight
Conceptual accuracy & correct physics	40%
Numerical work — correct setup, units, final answers	30%
Originality & depth of independent reasoning	20%
Clarity, organization & citation (where applicable)	10%

Declaration

I confirm that this submission represents my own original work, written in my own words, and has not been copied from another student, an online source, or an AI tool.

Name: _____ Roll/Seat No.: _____

Section: _____ Date: _____

A document sample is given to demonstrate how your document should be arranged…